

Group 25 Final Progress Report: Language Classification of Various Texts

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1 Introduction

Language Classification of Various Texts: Training a natural language processing (NLP) model to identify the language of origin for a given piece of text, being able to differentiate between English (en), Spanish (es), Italian (it), Afrikaans (af), Portuguese (pt), French (fr), Dutch (nl), German (de), Turkish (tr), Indonesian (id), Swedish (sv), and Tagalog (ta). With challenges related to it being difficult to differentiate between two languages, such as Spanish (es) and Portuguese (pt), resulting from subtle differences and similar words. The aim is to help linguistic experts quickly and easily identify the language of origin of texts, which can directly aid in the tasks of translation and understanding.

2 Related Work

The task of language classification has been performed multiple times. Various papers such as the *MMTEB: Massive Multilingual Text Embedding Benchmark* outlined a community project that covers over 250 languages using more than 500 quality control evaluation tasks (Enevoldsen et al., 2025). With the dataset that was selected from Amazon's MASSIVE dataset which contains more than 50 languages, projects such as the *MTEB: Massive Text Embedding Benchmark* spans 8 embedding tasks covering a total of 58 datasets and 112 languages, which exceeds the number of overall languages in this dataset (FitzGerald, 2025; Muennighoff et al., 2023). Muennighoff et al. determined that no single model dominates, as they all seem to converge to a similar efficacy rate. However, one point that multiple sources, including the paper from Fitzgerald states that the models trained on high-resource languages such as English, but performs significantly worse on low-resource languages (FitzGerald et al., 2022).

3 Dataset

The dataset used in this project is taken from the [Amazon Massive Scenario Classification dataset published by Massive Text Embedding Benchmark \(MTEB\)](#) through Hugging Face. It is a collection of sentences and/or commands that are transcribed from Amazon's Alexa devices. These transcriptions are all in English and are then professionally translated into 50 other languages. Thus, each entry in the dataset is an utterance or a spoken sentence of a user. The complete dataset includes 51 languages however, we will only be using 12 out of the 51 languages and a subset of the 15000 total instances for each of the languages in this project.

In the end our data was split as follows; training dataset consists of 5000 unique points per language for a total of 60,000 points, validation dataset consists of 24,396 total points, and finally testing consists of 35,688 points.

The data is then preprocessed using model specific tokenizers before we performed fine-tuning.

4 Features and Inputs

The original dataset consists of 5 columns: id, label, label_text, text, and lang. The column 'id' acts as a numerical identification system for the points and thus has no semantic value or information otherwise. This column would not be used for training our models. The columns 'label' and 'label_text' are redundant as the label information is equal to the label_text and they again, do not provide any semantic value for training and were removed when we were creating our datasets. The column 'lang' is the expected label of each point and should not be used for training. This leaves us with the column 'text'. This was the only column that would be used for training and was thus the only way one could derive any features or input values for the models. This means that the input features would consist solely of tokenized text as word embed-

dings.

5 Implementation

For the final step of the project, a total of four model implementations were tested on the language classification of various texts dataset: Flan-T5 (google/flan-t5-base), BERT (bert-base-uncased), mDeBERTa-V3(microsoft/mdeberta-v3-base), and mmBERT(jhu-clsp/mmBERT-base). The performance of these models on the test split of the dataset was compared against the baseline models: Random Baseline, and BoW + Logistic Classifier.

The complete performance results of all the models can be found in Table 7.1.

The initialization and loss functions for all models, apart from the baseline, were identical, with the optimization steps being fundamentally the same with small alterations in the hyperparameter values to be suit each individual model.

All models and their respective tokenizers were initialized by being directly imported from HuggingFace through the use of either their respective libraries or the universal `AutoModelForSequenceClassification` and `AutoTokenizer` libraries. Each model's respective tokenizer was used for tokenizing the test and validation dataset splits when that model was being run.

The main performance metrics being calculated for all the models that were calculated during training are the training accuracy and F1 score, with the training loss and validation loss being calculated at the end of every epoch.

The optimization steps for all the non-baseline models mainly composed of using LoRA adapter layers, which froze majority of the layers for all the models, only leaving the LoRA layers and the classifier head unfrozen so that the entire model is not being retrained, which is very large and would take a lot of computation power and time, but instead only training the classifier head of the model which adapts it to the use case of language classification of various texts. A breakdown for the specific hyperparameter values as well as why each specific model was chosen is as follows.

5.1 Flan-T5

Unlike the BERT variants, the Flan-T5 model is an encoder-decoder model. Using this model

provided us with an opportunity to see how different model architectures would perform on the same task. The model was trained on numerous different datasets and tasks, 473 datasets, 146 task categories, and 1,836 total tasks to be exact, in many different languages. The base model was chosen over the larger models for ease of running on a CPU. In the case of the experiment, the Flan-T5-base model was initialized with the following: `batch_size=32`, `learning_rate=1e-4`, `epochs=5`, `weight_decay=0.01`, and `dropout=0.0`.

5.2 BERT

BERT base uncased was selected since it is one of the most basic English models, which offers a highly versatile, open-source pre-trained model that achieves excellent performance on various NLP tasks (classification, tagging, Q&A) while remaining light enough to fine-tune on a single consumer GPU. The model is trained on BookCorpus, a dataset consisting of 11,038 unpublished books and English Wikipedia. For the task, BERT base uncased provides an interesting lens into whether a model that was trained only on English sources can accurately and effectively identify the languages. In the case of the experiment, the BERT model was initialized with the following: `batch_size=16`, `learning_rate=0.005`, `epochs=5`, `weight_decay=0.001`, and `dropout=0.0`.

5.3 mDeBERTa-V3

mDeBERTa was selected due to 2 reasons. The first being that DeBERTa is an improved version of BERT and RoBERTa and the other is that the "m-" in mDeBERTa stands for multilingual. It is fine-tuned on the CC100 multilingual dataset on top of the dataset that was used to train the base DeBERTa model, which is very similar to what is used for BERT and RoBERTa. For this task, the mDeBERTa-V3 model was initialized with the following: `batch_size=16`, `learning_rate=2e-5`, `epochs=3`, `weight_decay=0.001`, and `dropout=0.0`.

5.4 mmBERT

This model was selected because it is fundamentally the ModernBERT architecture, but specifically trained on over three trillion tokens across 1800 languages. It has the power of the refined ModernBERT model, which is an

Model Name	F1-Score	Accuracy	Precision	Recall
Random Baseline	0.08	0.08	0.08	0.08
BoW + Logistic Classifier	0.97	0.97	0.97	0.97
Flan-T5	0.97	0.97	0.97	0.97
BERT	0.97	0.97	0.97	0.97
mDeBERTa-V3	0.96	0.96	0.96	0.96
mmBERT	0.98	0.98	0.98	0.98

Table 5.1: Performance of the group 25 models on test dataset.

improved version of the original BERT model, and was specifically designed to be a modern multilingual encoder built for being able to work with various languages. For the task of language classification through various texts, this model had the great potential to perform very well, and as the received results are observed, it did perform well, beating all of the other selected models. In the case of this experiment, the mmBERT model was initialized with the following: `batch_size=16`, `learning_rate=2e-5`, `epochs=3`, `weight_decay=0.001`, and `dropout=0.0`.

6 Results and Evaluation

6.1 Baselines

The simple baseline that was selected was the *random baseline*. Since there are 12 classes, this would result in the accuracy being around $\frac{1}{12}$ or around 0.08. The other simple baseline that could have been used, *majority class*, would have yielded similar results since all the classes were balanced.

For the other baseline, an implementation of BoW + Logistic Classifier was used. The results of this implementation was significantly better, hitting 0.97 for F1-score, accuracy, precision, and recall. This was considered the main baseline to beat.

6.2 Group Models

The members of group 25 used four different models to compete against the baseline. These models were listed in section 5 and are the following:

- Flan-T5
- BERT (BERT base uncased)
- mDeBERTa-V3
- mmBERT

All the models used a LORA-adaptor and used their corresponding preprocessing and tokenizers.

While the first three performed marginally better compared to the Logistic Classifier Baseline, mmBERT performed a whole percent better. One point of note is that all the models were able to properly identify the languages in the sanity check, whereas the Logistic classifier messed up a few of the statements.

Below is the sanity check:

1. jangan pergi main di sana
2. what is your name?
3. escuela de idiomas
4. je suis un étudiant
5. eetsmakelijk
6. buon appetito
7. danke schön

Statement 5 and 7 were identified incorrectly in the baseline, as statement 5 means "enjoy your meal" in Dutch (misclassified as Turkish in baseline) while statement 7 means "thank you very much" in German (misclassified as Turkish in baseline).

6.3 Peer Models

Various models used by peers performing the Group 25 Codabench used BERT, DistilBERT, and RoBERTa models. For the various BERT models, they performed similarly, hovering around a 0.96-0.97 accuracy, precision, recall, and F1-score. The same can be said for the DistilBERT and RoBERTa models. For all the models, they used a LORA-adaptor.

7 Comparison to Others

To summarize the results, only the best performing version of each corresponding model will be listed. In addition to this, zero-shot model results will not be listed as to remove clutter and since

Model Name	F1-Score	Accuracy	Precision	Recall
Majority Baseline	0.08	0.08	0.08	0.08
Random Baseline	0.08	0.08	0.08	0.08
BoW + Logistic Classifier	0.97	0.97	0.97	0.97
RoBERTa (Peer)	0.93	0.93	0.93	0.93
DistilBERT (Peer)	0.96	0.96	0.96	0.96
Flan-T5 (Group 25)	0.97	0.97	0.97	0.97
mDeBERTa-V3 (Group 25)	0.96	0.96	0.96	0.96
mmBERT (Group 25)	0.98	0.98	0.98	0.98
BERT (Peer)	0.98	0.98	0.98	0.98
DistilBERT Multilingual (Peer)	0.99	0.98	0.98	0.98

Table 7.1: Best performing models on the test dataset.

they performed significantly worse compared to their fine-tuned model counterparts.

Most of the models used around 1-5 epochs depending on their machine, with varying learning rates from 0.001 to 0.005. One note is that for the DistilBERT Multilingual model, the peer did not use an adapter and just trained for 5 epoches, whereas the other ones all used the LORA adapter.

The results are pretty on par with what was expected and is in line with the models group 24 trained. The dataset provided an even distribution for the various languages, which allowed for minimal skewing in terms of language. The minor variations in results other than model selection can be attributed to different hyperparameters and differing epochs. During training, training with one epoch already yielded an accuracy of 0.97 for the validation set.

8 Error Analysis

As mentioned in section 6.2, a sanity check was performed at the end to identify whether the metrics seem reasonable. By doing this, we were able to see that the baseline models often fail to identify phrases that consists of words that are not a part of the training dataset. In comparison, all of the fine-tuned models that were trained by our group members were able to identify these phrases correctly. I suspect that while the logistic regression model only relies on the dataset given to make the predictions, the fine-tuned models have more context on words that are not a part of the given dataset. Thus, it was able to make a more accurate prediction.

Furthermore, we noticed that while models (except for the baselines) give an impressive accuracy number, they have a few quirks that are worth

noting. Firstly, the model seems to struggle with classifying either very short or very long sentences. The other quirk is that due to the nature of the task being a single label task, each of the model that we trained seems to classify words that are common to more than one language differently. For example, the word "gratis" that means "free" in many languages are classified as different languages by different models. The idea that some words can be shared between multiple languages means that it is possible that some of the misclassified instances were not misclassified, but were actually inaccurately labelled.

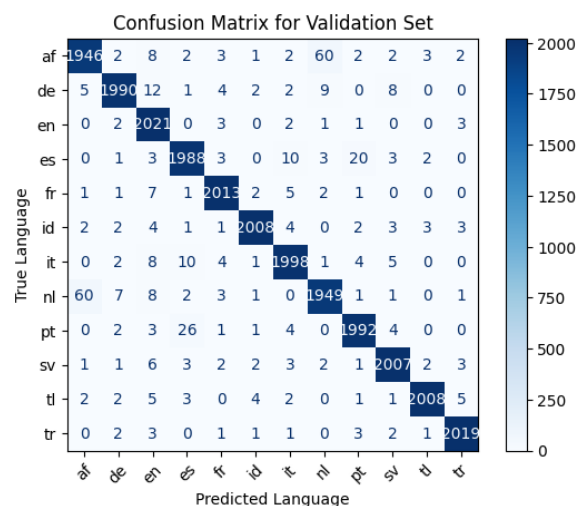


Figure 8.1: Confusion matrix for the best-performing model by our group (mmBERT).

Moreover, if we do a quantitative analysis on the best model that one of our group members have made by creating a confusion matrix, we observe that the bulk of the error comes from the model mistaking the Spanish-Portuguese pair and the Dutch-Afrikaans pair. As seen in figure 8.1,

there are 120 misclassified instances of the Dutch-Afrikaans pair and 46 instances of the Spanish-Portuguese pair. This same pattern is also seen in the other models that our group have fine-tuned. Thus, it confirms our hypothesis from the annotation project step that these two language pairs are the hardest to distinguish from one another.

9 Conclusions

The task of classifying languages seem simple and straightforward, but there can be some quirks and nuances that we have identified while working on this task by looking beyond the metrics. As mentioned in section 8, languages that share a common word often prove to be challenging for the models, especially when given only the word itself. Furthermore, we only tried the LoRA adapter for this task, which limits the number of parameters our model was able to learn. As shown by one of our peers, if we tried a different adapter method or electing to train the model without using an adapter, it may provide some improvements to the model. We also note that the models that were pre-trained on multilingual dataset tends to outperform the ones that were only pre-trained on English datasets.

Ultimately, given the time and resources to improve the model, we would modify this task into a multi-label language classification task and split up the phrases into words which would have multiple correct labels. Then, we would compile a set of models that are pre-trained on multilingual datasets and a set of adapters to perform a comparison between all the model-adapter-hyperparameter combinations. We believe that by doing this, the models would be able to more accurately classify the languages despite the similarity of some languages or the length of the input.

10 Team Contributions

10.1 Elite

- Implemented the BERT model.
- Wrote sections 2, 6, & 7.

10.2 Tania

- Implemented the Flan-T5 Model.
- Wrote sections 3 & 4

10.3 Kenneth

- Implemented the mDeBERTa-V3 Model.
- Wrote sections 8 & 9

10.4 Ishpreet

- Implemented the mmBERT Model.
- Wrote sections 1 & 5

References

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